

ValueError: 'to_matrix' will return an exponentially large matrix

<https://github.com/qiskit-community/qiskit-optimization/issues/401>

ROW 47

ValueError: 'to_matrix' will return an exponentially large matrix #401

New issue



Closed



gcattan opened on Jul 19, 2022

...

Environment

- Qiskit Optimization version: 0.4.0
- Python version: 3.8
- Operating system: Windows 10

What is happening?

I guess it is more a question than a real issue.

I have an optimization problem, with 5 variables.

Each variable is a matrix of size 2x2 containing integer variables.

I am using `IntegerToBinary` to convert the problem, as below:

```
def _solve_qp(self, qp):
    conv = IntegerToBinary()
    qubo = conv.convert(qp)
    backend = BasicAer.get_backend('statevector_simulator')
    quantum_instance = QuantumInstance(backend)
    qaoa_mes = QAOA(quantum_instance=quantum_instance,
                    initial_point=[0., 0.])
    qaoa = MinimumEigenOptimizer(qaoa_mes)
    return conv.interpret(qaoa.solve(qubo))
```



When I run the code above, I am getting this error:

```
ValueError: 'to_matrix' will return an exponentially large matrix, in this case '7588550360256754183279'?
```



I am not sure whether my reasoning is wrong or if this is a limitation of Qiskit.
Could you help me, please?

How can we reproduce the issue?

N/A

What should happen?

N/A

Any suggestions?

No response



t-imamichi on Jul 20, 2022

Collaborator

...

Could you paste the entire script to [gist](#) to reproduce the error?



gcattan on Jul 20, 2022

Author

...

Sure. Please find the gist [here](#).



Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants





gcattan on Jul 27, 2022

Author ...

@t-imamichi were you able to reproduce the error with the gist?



t-imamichi on Jul 28, 2022 · edited by t-imamichi

Edits Collaborator ...

Because you use unbound integer variables (no lower bound and no upper bound), they are converted into a large set of binary variables to generate a qubo. I printed out your problem and the result is as follows. The number of variables (272) is too much, which corresponds to the number of qubits.

I suggest setting appropriate bounds for the integer variables.

```
input problem Problem name: docplex_model1

Minimize
  5*fro_mean_0_0^2 + 5*fro_mean_0_1^2 + 5*fro_mean_1_0^2 + 5*fro_mean_1_1^2

Subject to
  No constraints

Integer variables (4)
  fro_mean_0_0 # no lower bound and no upper bound displayed.
  fro_mean_0_1
  fro_mean_1_0
  fro_mean_1_1

num variables of qubo: 272
```



You defined the variables as follows. `infinity` should not be used.

```
X_mean = prob.integer_var_matrix(keys1=channels, keys2=channels, name="fro_mean", lb=-prob.infinity)
```



gcattan on Jul 29, 2022

Author ...

Ok, I understand, thank you for your answer.



gcattan closed this as completed on Jul 29, 2022